Yoonho Kim

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Education

University of Toronto Toronto, ON
BASc in Engineering Science - Robotics Engineering, Minor in AI *Exp. May 2027*

Experience

Software Engineer Intern May 2026 – Aug 2026
Index Exchange *Toronto, ON*

- Built and launched the core API layer for an identity data platform in Golang, delivering 8 new REST endpoints that gave users more autonomy over segment data and successfully scaled to 20,000+ daily production syncs
- Engineered a real-time event-driven pipeline within the platform using Golang, Kafka, and Aerospike to instantly stream and persist user identity changes in near real time
- Optimized the platform's RTB auction path to enrich bid requests across all user identifier types, successfully supporting a peak volume of 30,000 identity lookups per minute
- Generalized audience-segment associations across internal API, file upload, and partner endpoint ingestion flows, consolidating previously separate paths into a single association model, significantly reducing technical debt

Software Engineer Intern Sept 2025 – April 2026
Auvik Networks *Toronto, ON*

- Developed and maintained Go-based microservices processing millions of device records using Kafka, PostgreSQL, Debezium CDC, and Kubernetes
- Led large-scale backend migration refactoring 15K+ LOC, replacing callback-driven iterators with optimized PostgreSQL batch queries, reducing query costs by 90% and cutting service-level memory usage by 24%
- Optimized complex PostgreSQL queries, eliminating production OOM failures and reducing execution latency by up to 96%
- Designed thread-safe test datastore and reusable test utilities, reducing test runtime by 40% across 30+ integration test suites
- Modernized legacy services by removing 9.8K+ lines of deprecated code and migrating all integration tests to a unified PostgreSQL-backed persistence layer

Software Engineer Intern Sept 2024 – Aug 2025
SEW Eurodrive of Canada *Toronto, ON*

- Built a real-time AGV communication system using multi-threaded Python and UDP, reducing pickup execution time by 50%
- Implemented a 3D perception pipeline using ifm3dpy, OpenCV, and PCL, achieving 80%+ object detection accuracy in production environments
- Designed and deployed 8 REST/WebSocket API endpoints for issuing navigation commands and telemetry ingestion across multiple vehicles
- Built a React-based fleet management dashboard, reducing manual monitoring effort by 40% through real-time visualization of vehicle state, routing, and operational metrics

Projects

TaskBolt | *TypeScript, Fastify, PostgreSQL, Prisma, Redis, Docker, OAuth*

- Built a task management platform supporting hierarchical subtasks, recurring schedules, and Redis-backed reminders
- Implemented bidirectional Google Tasks API synchronization with conflict resolution
- Integrated LLMs to convert natural language into structured tasks with 85%+ task level correctness

Technical Skills

Languages: Python, C++, C, Golang (Go), JavaScript/TypeScript, SQL, HTML/CSS
Frameworks/Libraries: React, Node.js, Fastify, Django, Flask, PyTorch, OpenCV, NumPy, pandas, TailwindCSS
Data & Infrastructure: PostgreSQL, Kafka, Redis, AWS, Aerospike, Debezium, Prisma
Developer Tools: Docker, Kubernetes, Grafana, Datadog, Git, Linux, VS Code, Cursor, Claude Code, Codex